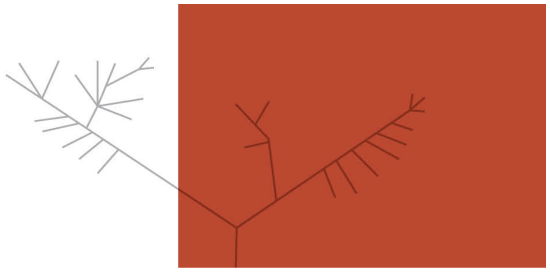
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Special Methods for Detecting Antimicrobial Resistance Part 2 [ESBL, Carbapenemase test]

AST Workshop – Lecture 17

Vlademir Cantarelli, PhD

What is an ESBL?

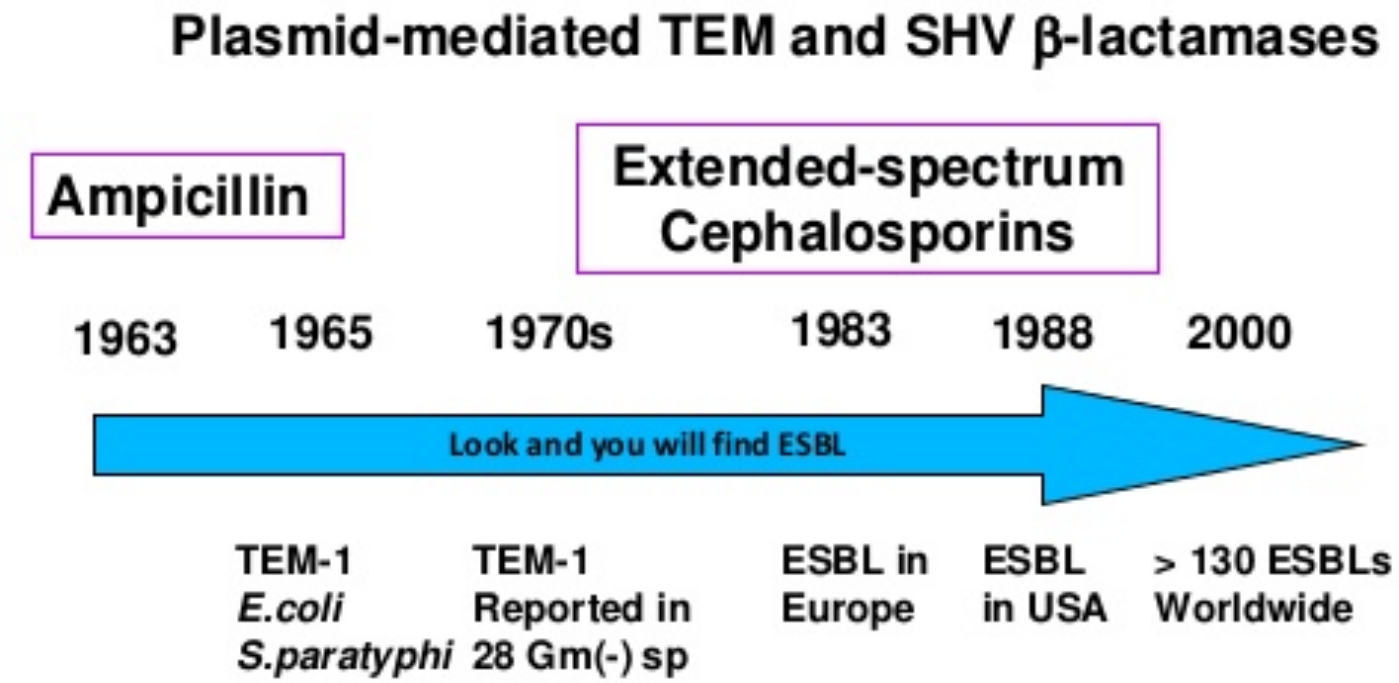


- **Extended Spectrum Beta lactamases**
  - enzymes that mediate resistance to extended-spectrum (third generation) cephalosporins (e.g., ceftazidime, cefotaxime, and ceftriaxone) and monobactams (e.g., aztreonam) but do not affect cephamycins (e.g., cefoxitin and cefotetan) or carbapenems (e.g., meropenem or imipenem).
  - Can result in treatment failure if one of the above drugs are used to treat an infection.

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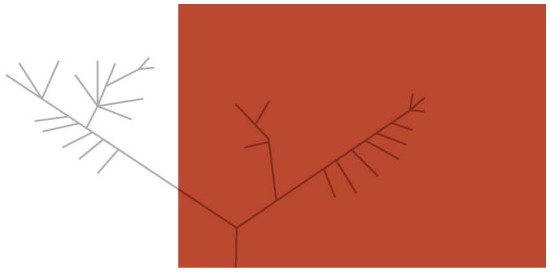
Evolution of Beta-Lactamases





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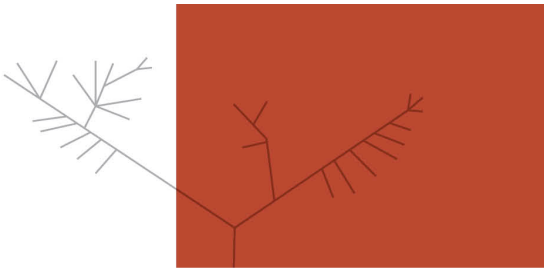
ESBL detection



- ESBL Tests
  - Since CLSI M100-S20:
    - Routine ESBL testing **NOT** necessary if currently available interpretative criteria are used by the lab
    - **No need** to edit results for cephalosporins, aztreonam and penicillins from susceptible to resistant
  - ESBL confirmation may be useful for epidemiological or infection control purposes

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Special Methods



- ESBL Tests
  - When to perform?

	Antimicrobial disk [ ]	Confirm ESBL if ≤	Or MIC ≥
E. coli *K. pneumoniae* *K. oxytoca*	Cefpodoxime 10 mg	17 mm or	8 µg/ml or
	Ceftazidime 30 mg	22 mm or	2 µg/ml or
	Aztreonam 30 mg	27 mm or	2 µg/ml or
	Cefotaxime 30 mg	27 mm or	2 µg/ml or
	Ceftriaxone 30 mg	25 mm	2 µg/ml
Proteus mirabilis	Cefpodoxime 10 mg	22 mm or	2 µg/ml or
	Ceftazidime 30 mg	22 mm or	2 µg/ml or
	Cefotaxime 30 mg	27 mm	2 µg/ml

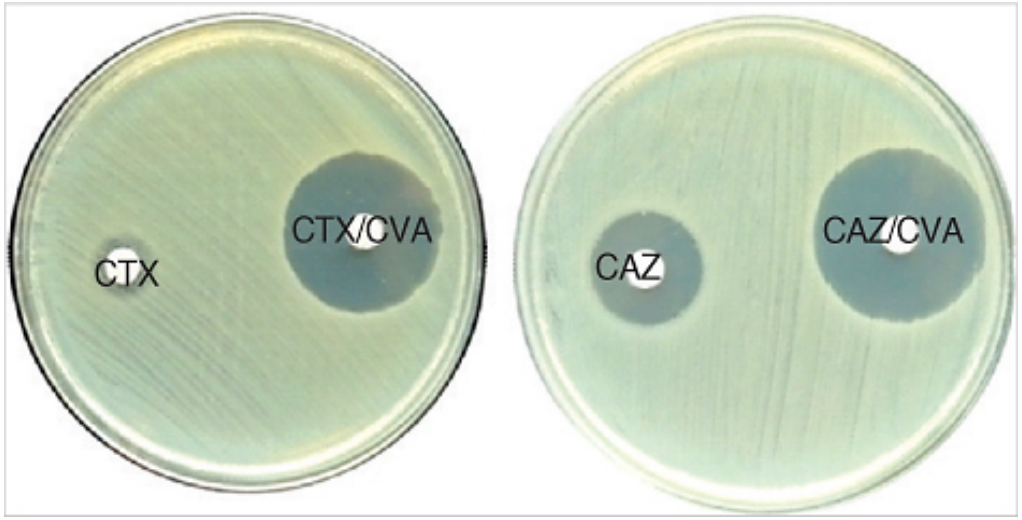
Standard disk diffusion or broth dilution procedure

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Special Methods

- ESBL Tests
  - How to perform?

Cefotaxime 30 μg
Cefotaxime + clavulanate 30/10 μg



Ceftazidime 30 μg
Ceftazidime + clavulanate 30/10 μg

Korean J Lab Med. 2008

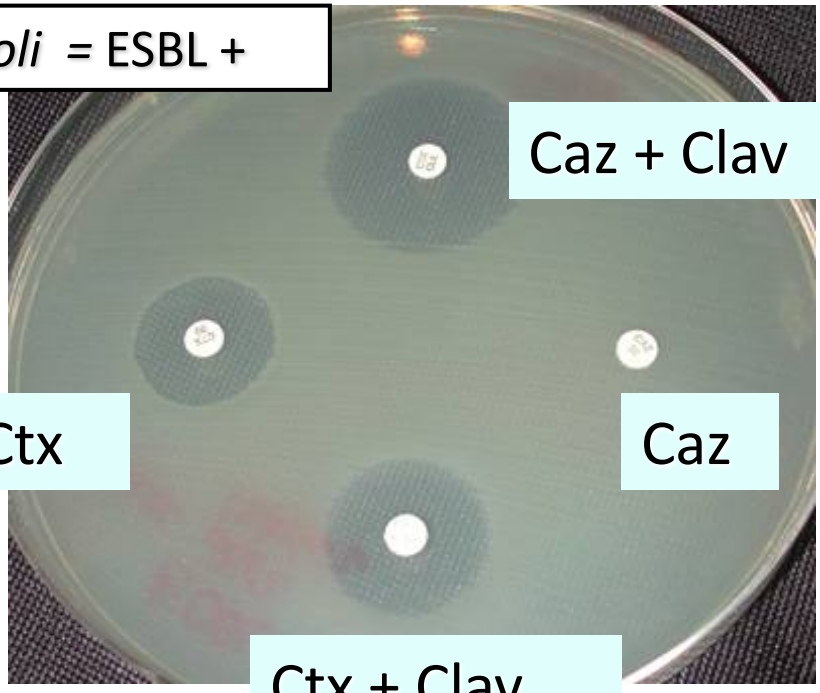
Interpretation:
A ≥5-mm increase in a zone diameter for either antimicrobial agent tested in combination with clavulanate vs the zone diameter of the agent when tested alone = ESBL

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Detection of ESBLs

- Use 4 disks per organism
- ≥5 mm larger zone size with disk + Clavulanate than same drug without Clavulanate = ESBL

E. coli = ESBL +



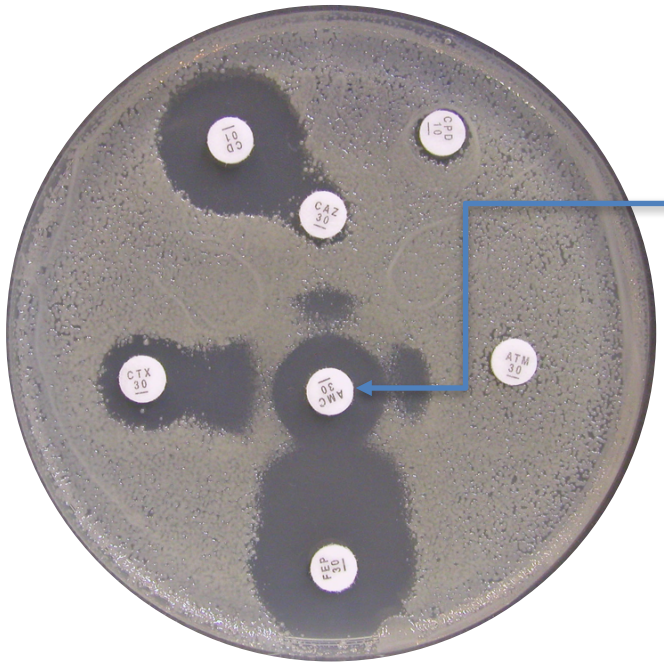
K. oxytoca = Not ESBL



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Detection of ESBLs

- Disk approximation test
  - How to perform? (Not recommended by CLSI)

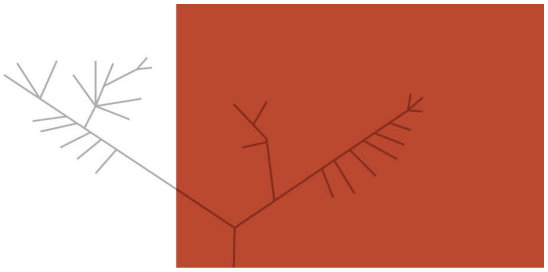


EUCAST

Interpretation:
A positive result is indicated when the inhibition zones around any of the cephalosporin disks are augmented in the direction of the disk containing clavulanic acid

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Special Methods



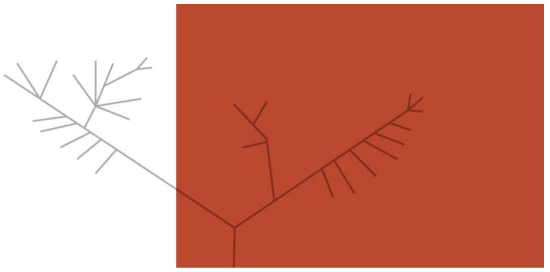
- ESBL QC recommendation

ATCC	Antimicrobial disk [ ]	Zones (mm)	Or MIC ≥
K. pneumoniae 700603	Cefpodoxime 10 mg	9 – 16	8 µg/ml
	Ceftazidime 30 mg	10 – 18	2 µg/ml
	Aztreonam 30 mg	9 – 17	2 µg/ml
	Cefotaxime 30 mg	17 – 25	2 µg/ml
	Ceftriaxone 30 mg	16 – 24	2 µg/ml

Standard disk diffusion or broth dilution procedure

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Carbapenemases

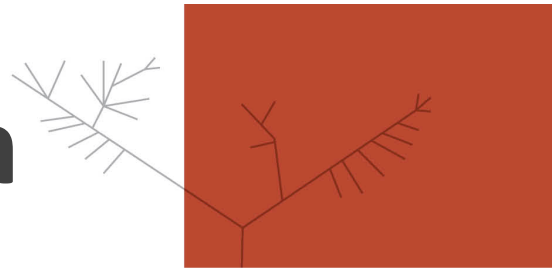


- Broad spectrum beta-lactamases able to hydrolyze:
  - penicillins, cephalosporins, monobactams and **carbapenems**

Enzyme	Location	Year of Isolation	Country of origin
SME	Chromosome	1982	London, UK
IMI	Chromosome	1984	California, USA
NMC-A	Chromosome	1990	Paris, France
IMP	Plasmid	1994	Japan
KPC	**Plasmid**	**1996**	**North Carolina, USA**
VIM	Plasmid	1997	Verona, Italy
OXA-48	**Plasmid**	**2003**	**Turkey**
NDM	**Plasmid**	**2010 (2002)**	**India**
OXA-181	**Plasmid**	**2011**	**India**

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Carbapenemase detection



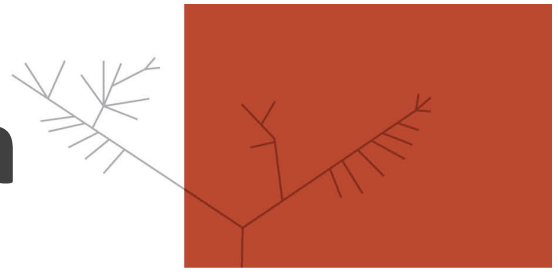
- For *Enterobacteriaceae*
  - No need to perform confirmatory test if current breakpoints (CLSI M100-S27) is implemented
    - Except for epidemiological or infection control purposes

	Disc Diffusion (10 μg)			Broth dilution (MIC)		
	S	**I**	**R**	**S**	**I**	**R**
Doripenem	≥23	20 – 22	≤19	≤1	2	≥4
Ertapenem	≥22	19 – 21	≤18	≤0.5	1	≥2
Imipenem	≥23	20 – 22	≤19	≤1	2	≥4
Meropenem	≥23	20 – 22	≤19	≤1	2	≥4

Breakpoints based on a specific dosage regimen

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Carbapenemase detection



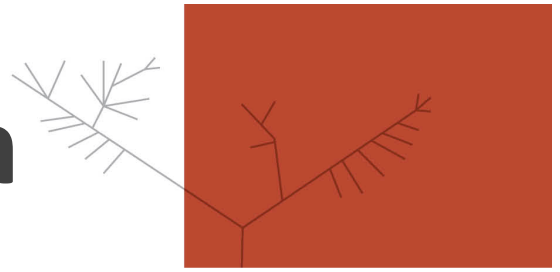
- For *Pseudomonas aeruginosa*
  - No need to perform confirmatory test if current breakpoints (CLSI M100-S27) is implemented
    - Except for epidemiological or infection control purposes

	Disc Diffusion (10 μg)			Broth dilution (MIC)		
	S	**I**	**R**	**S**	**I**	**R**
Doripenem	≥19	16 – 18	≤15	≤2	4	≥8
Imipenem	≥19	16 – 18	≤15	≤2	4	≥8
Meropenem	≥19	16 – 18	≤15	≤2	4	≥8

Breakpoints based on a specific dosage regimen

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Carbapenemase detection



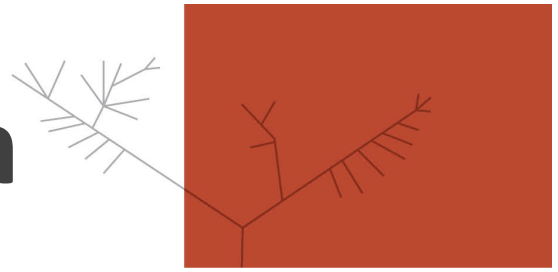
- For *Acinetobacter*
  - No need to perform confirmatory test if current breakpoints (CLSI M100-S27) is implemented
    - Except for epidemiological or infection control purposes

	Disc Diffusion (10 μg)			Broth dilution (MIC)		
	S	**I**	**R**	**S**	**I**	**R**
Doripenem	≥18	15 – 17	≤14	≤2	4	≥8
Imipenem	≥22	19 – 21	≤18	≤2	4	≥8
Meropenem	≥18	15 – 17	≤14	≤2	4	≥8

Breakpoints based on a specific dosage regimen

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Carbapenemase detection



Introduction to Tables 3B, 3C, and 3D. (Continued)

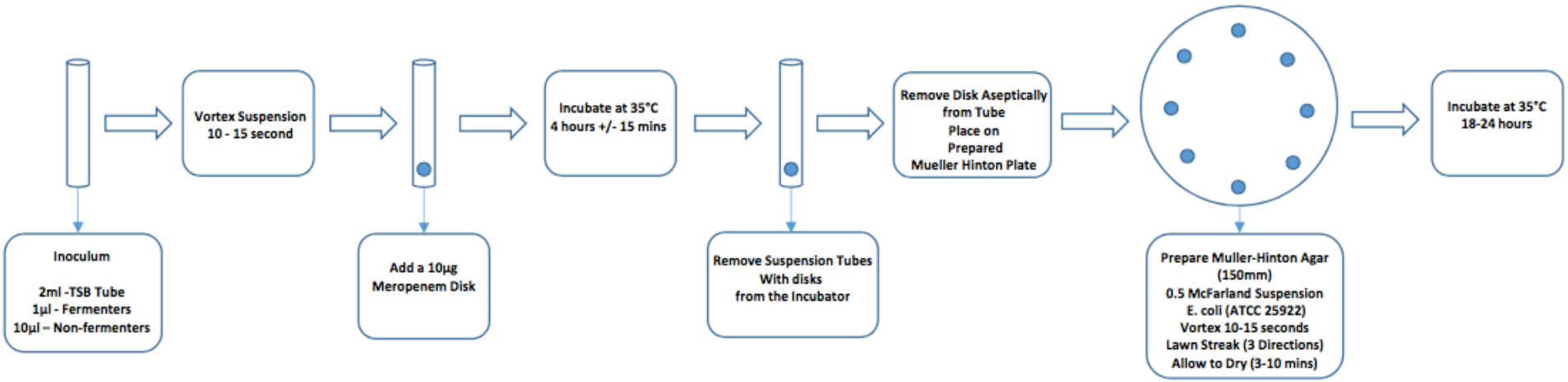
	Tests Used for Epidemiological or Infection Control–Related Testing			
	MHT (Table 3B)	**Carba NP (Table 3C)**	**mCIM (Table 3D)**	**Other (eg, molecular assays)**
Organisms	*Enterobacteriaceae* that are not susceptible to one or more carbapenems	*Enterobacteriaceae, P. aeruginosa,* and *Acinetobacter* spp. that are not susceptible to one or more carbapenems	*Enterobacteriaceae* that are not susceptible to one or more carbapenems	*Enterobacteriaceae, P. aeruginosa,* and *Acinetobacter* spp. that are not susceptible to one or more carbapenems to determine the presence of a carbapenemase, or to determine carbapenemase type in isolates positive by MHT or Carba NP
Strengths	Simple to perform No special reagents or media necessary	Rapid	No special reagents or media necessary	Determines type of carbapenemase in addition to absence or presence of the enzyme
Limitations	False-positive results can occur in isolates that produce ESBL or AmpC enzymes coupled with porin loss. False-negative results are occasionally noted (eg, some isolates producing NDM carbapenemase). Only applies to *Enterobacteriaceae*.	Special reagents are needed, some of which necessitate in-house preparation (and have a short shelf life). Invalid results occur with some isolates. Certain carbapenemase types (eg, OXA-type, chromosomally encoded) are not consistently detected.	Only applies to *Enterobacteriaceae* Requires overnight incubation	Special reagents and equipment are needed. Specific to targeted genes; false-negative result if specific carbapenemase gene present is not targeted.

Abbreviations: ESBL, extended-spectrum β-lactamase; MIC, minimal inhibitory concentration; MHT, modified Hodge test; NDM, New Delhi metallo-β-lactamase.

CLSI M100-S27

Special Methods: mCIM

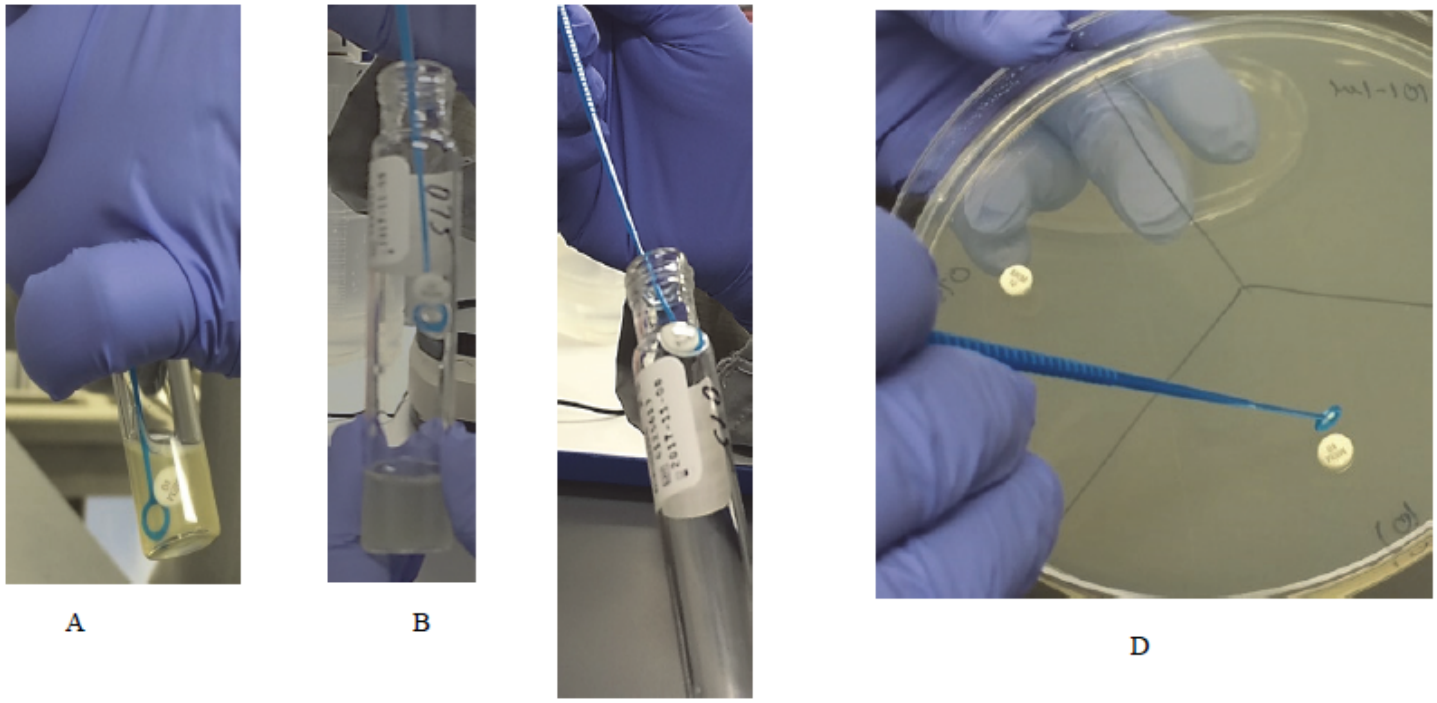
Modified Carbapenem Inactivation Method



For epidemiological or Infection Contro purposes only
Not recommended for routine use
No change in the interpretation of carbapenem susceptibility results is necessary for mCIM positive isolates (CLSI M100-S27)

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mCIM Test



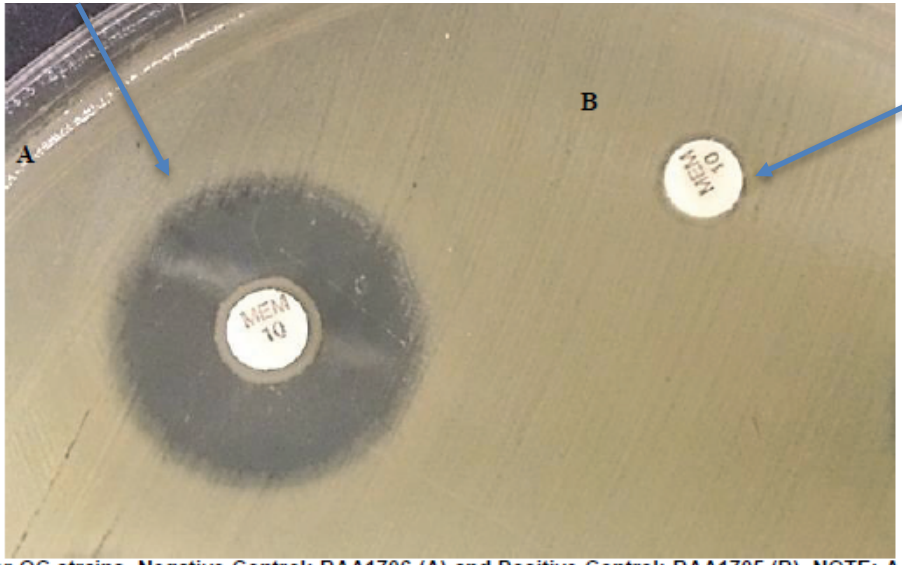
A B C D

Figure 1: Remove the meropenem disk with a 10µL loop (A) and drag the loop against the inside edge of the tube to expel any excess liquid (B). Use the same loop to remove the disk from the tube (C) and place it on the MHA plate (D) previously inoculated with the meropenem-susceptible *E. coli* (ATCC® 25922) indicator strain.

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mCIM Test

Carbapenemase
Negative: Zone ≥ 19 mm



Carbapenemase Positive:
Zone 6 – 15 mm or colonies within zone 6 – 18 mm

Zone 16 – 18 mm:
Indeterminate

Figure 2a: mCIM results for QC strains. Negative Control: BAA1706 (A) and Positive Control: BAA1705 (B). NOTE: A narrow ring of growth around the meropenem disk as seen with the negative control (A) results from carryover of the test organism in the TSB broth and should be ignored.



Figure 2b: Example of carbapenemase producing isolates. NOTE: Positive mCIM result.

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Quality control of mCIM

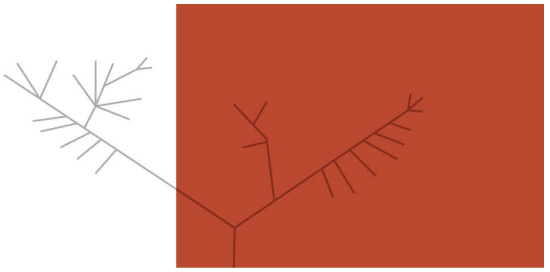
Test positive and negative QC strains each day of testing.

K. pneumoniae ATCC® BAA-1705™—Carbapenemase positive
K. pneumoniae ATCC® BAA-1706™—Carbapenemase negative

In addition, perform QC of meropenem disks and test media daily or weekly following the routine disk diffusion QC procedure and handle disks as described in CLSI document M02. Alternatively, perform QC of meropenem disks with each run by removing a disk from the cartridge of disks used for the run and placing it on the MHA plate inoculated with *E. coli* ATCC® 25922; incubate as above.

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mCIM Test Reporting



mCIM Result	Report as:
Positive	Carbapenemase detected
Negative	Carbapenemase not detected
Indeterminate	Testing inconclusive for the presence of carbapenemase. Call the laboratory for discussion

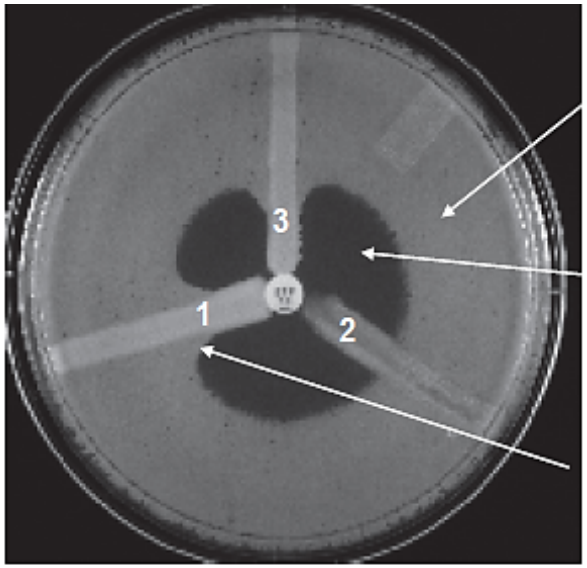
For indeterminate results:

- Check test isolate and *E. coli* 25922 for purity
- Check meropenem disc integrity by (routine QC results are acceptable)
- Repeat mCIM test for isolate and QC strain
- If repeat test is negative, consider confirmation by molecular testing

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Special Methods

- Carbapenemases in *Enterobacteriaceae*
  - *Modified Hodge Test- No longer recommended*



E. coli ATCC® 25922

Inhibition of *E. coli* ATCC® 25922 by ertapenem

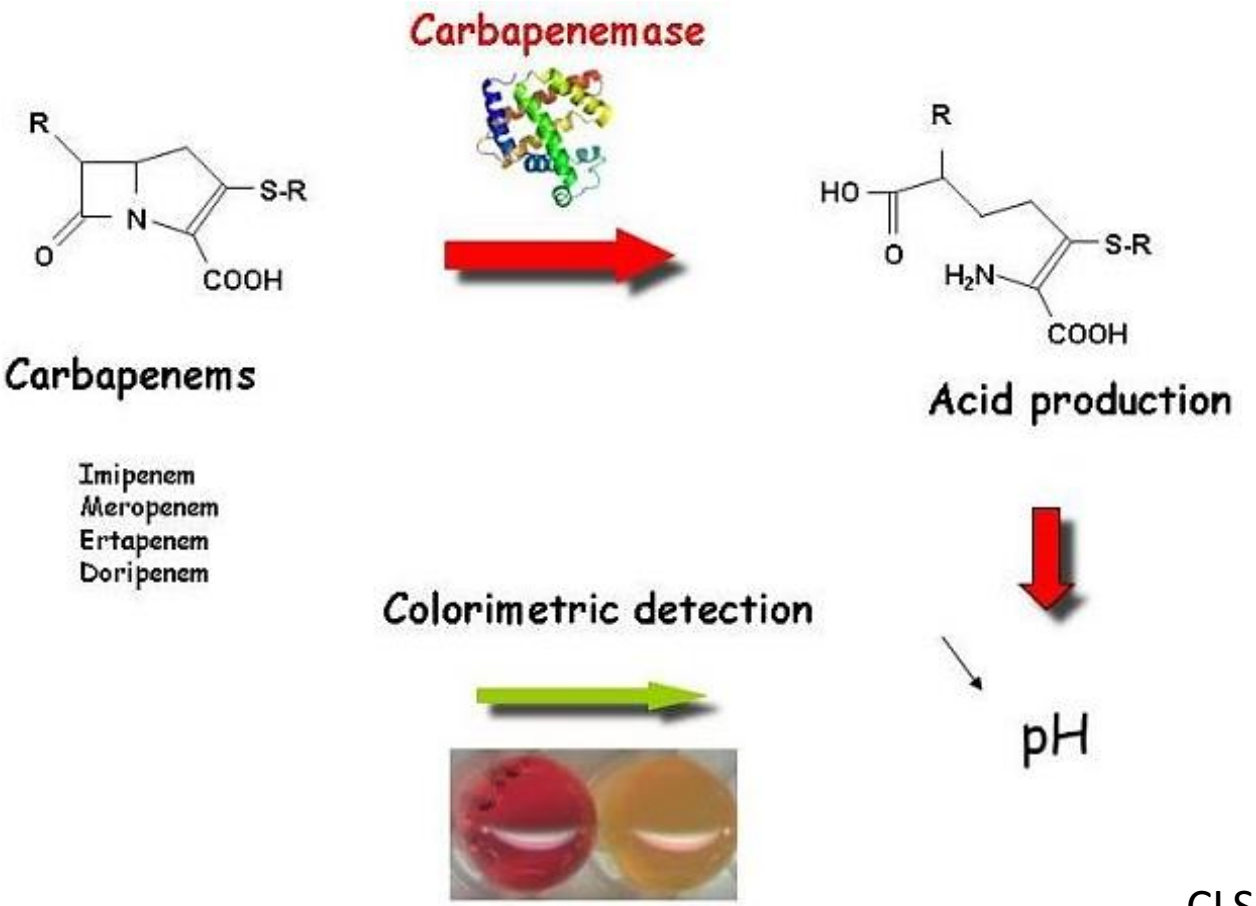
Enhanced growth of *E. coli* ATCC® 25922. Carbapenemase produced by *K. pneumoniae* ATCC® BAA-1705 inactivated ertapenem that diffused into the media. Thus, there is no longer sufficient ertapenem here to inhibit *E. coli* ATCC® 25922 and an indentation of the zone is noted.

Figure 1. The MHT Performed on a Small MHA Plate.
(1) *K. pneumoniae* ATCC®BAA-1705, positive result;
(2) *K. pneumoniae* ATCC®BAA-1706, negative result;
and (3) a clinical isolate, positive result.

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Special Methods

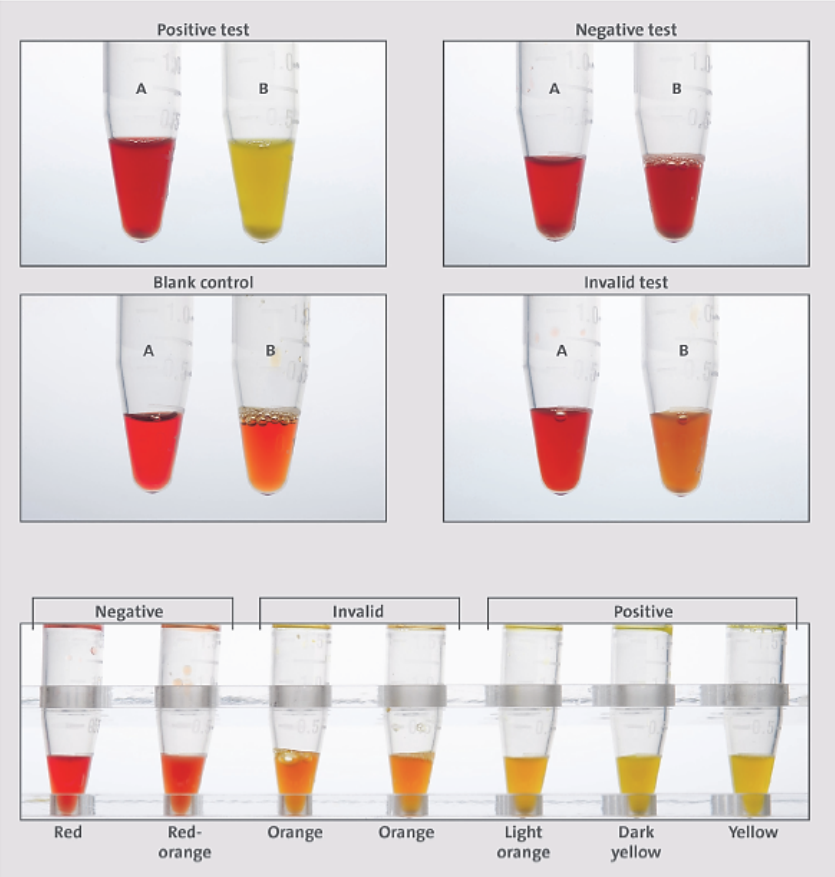
- Carbapenemases in *Acinetobacter*
  - *Carba NP Test*



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Special Methods

- Carbapenemases in *Acinetobacter*
  - *Carba NP Test interpretation:*



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CLSI M100S26E Page: 133

Special Methods

- Carbapenemases in *Enterobacteriacea*

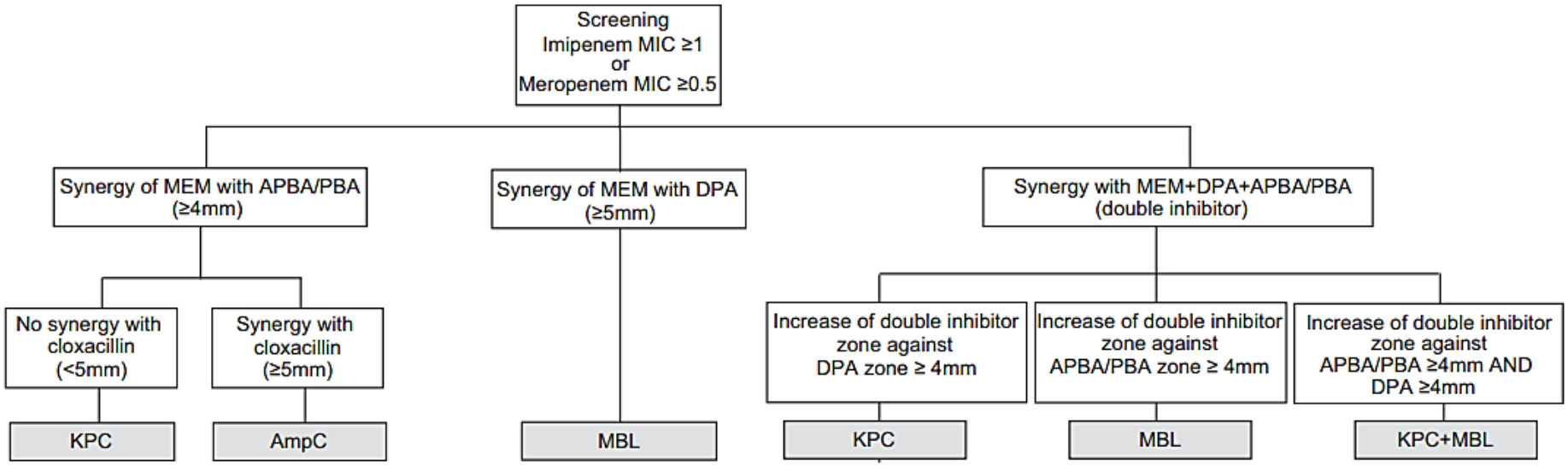


FIG. 1. Algorithm used for interpretation of results obtained by the KPC/Metallo-β-Lactamase Confirmation-based Methods A and B that included an additional double inhibitor disc.

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Special Methods

- How can we determine the type of carbapenemase involved?

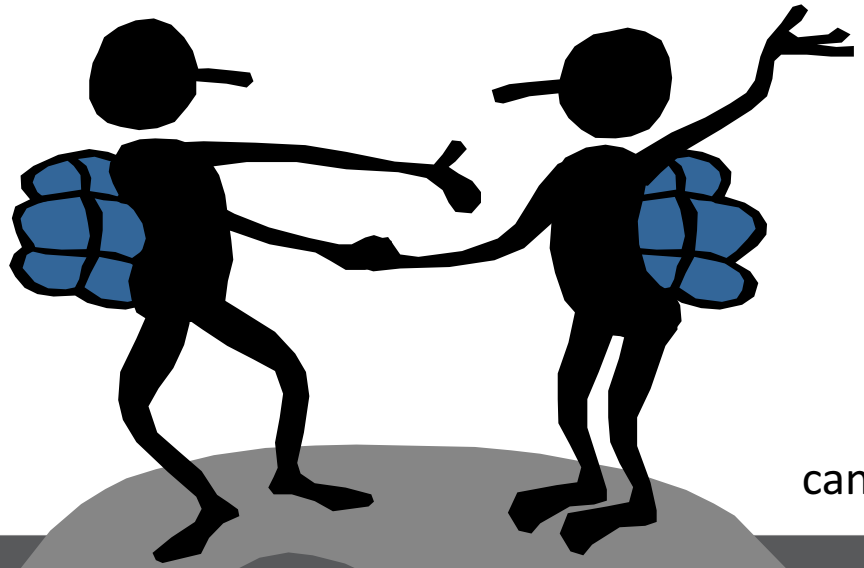


Molecular Methods

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Thank you!



cantarelliv@gmail.com